PROBLEM 18 | Problems T2 defects.pdf | Crystal Defects & Solid Solutions

Ranking of Solubility in Iron by the Hume-Rothery Rules

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> ***4. Using data in the table below, order the elements according to their solubility in iron, from higher to lower solubility. Justify your answer.*** *(Solution:* $\mathrm{Mo} > \mathrm{Ni} > \mathrm{Mn}$*) / Element / Atomic radius (nm) / Crystalline Structure / Electronegativity / Valence / / :*
>
> **Solution:** $\mathrm{Mo} > \mathrm{Ni} > \mathrm{Mn}$

PHASE 1: HYPOTHESIS & PARAMETERS

Solvent Matrix:

- **Iron** (α-Fe)**:**
- Atomic radius: $R_{\mathrm{Fe}} = 0.124$ nm.
- Crystal lattice: **BCC**.
- Pauling electronegativity: $\chi_{\mathrm{Fe}} = 1.7$.
- Stable valences: $+2, +3$.

Solutes to Evaluate:

1. **Nickel** (Ni)**:** $R = 0.125$ nm, FCC, $\chi = 1.8$, Val = $+2$.
2. **Molybdenum** (Mo)**:** $R = 0.136$ nm, BCC, $\chi = 1.3$, Val = $+3, +4, +6$.
3. **Manganese** (Mn)**:** $R = 0.112$ nm, Simple Cubic, $\chi = 1.6$, Val = $+2, +3, +6, +7$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Solid solubility in substitutional solutions is governed by the **4 Hume-Rothery rules** [Session 4 Slide 23]:

1. **Atomic radius difference (ΔR):** It must be below 15% for appreciable solubility:

$$\Delta R = \frac{|R_{\mathrm{solute}} - R_{\mathrm{Fe}}|}{R_{\mathrm{Fe}}} \times 100\%$$

2. **Crystal structure:** Solute and matrix must share the **same lattice structure**. If they differ (e.g. BCC vs FCC or SC), complete solubility is physically impossible.
3. **Electronegativity:** The difference $|\Delta\chi|$ must be minimal. Large differences favour intermetallic compounds instead of a solid solution.
4. **Valence:** Same valence, or a solute valence compatible with the oxidation states of the matrix.
